

Predictive Modeling of Water Quality for Potability

Course: CHEF315: Machine Learning for Chemical
Engineers

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Problem Statement & Motivation

Access to Safe Drinking Water

Problem

Access to safe drinking water is a critical global issue.

Current Method

Traditional laboratory chemical analysis is expensive and time-consuming.

Objective

Develop a Machine Learning model to automate the classification of water samples as Potable (1) or Not Potable (0).

Key Input Parameters

pH, Hardness, Solids, Chloramines, Sulfates, Conductivity, Organic Carbon, Trihalomethanes, Turbidity.

Data Sourcing & Preprocessing

1

Source

Publicly available dataset (Kaggle/UCI) containing 3000+ water samples.

2

Data Cleaning

Handled missing values (NaN) in pH and Sulfate columns using Mean Imputation.

3

Scaling

Applied StandardScaler to normalize features (ranging from pH ~7 to Solids ~20,000) for uniform model interpretation.

Split

80% Training Data, 20% Testing Data.

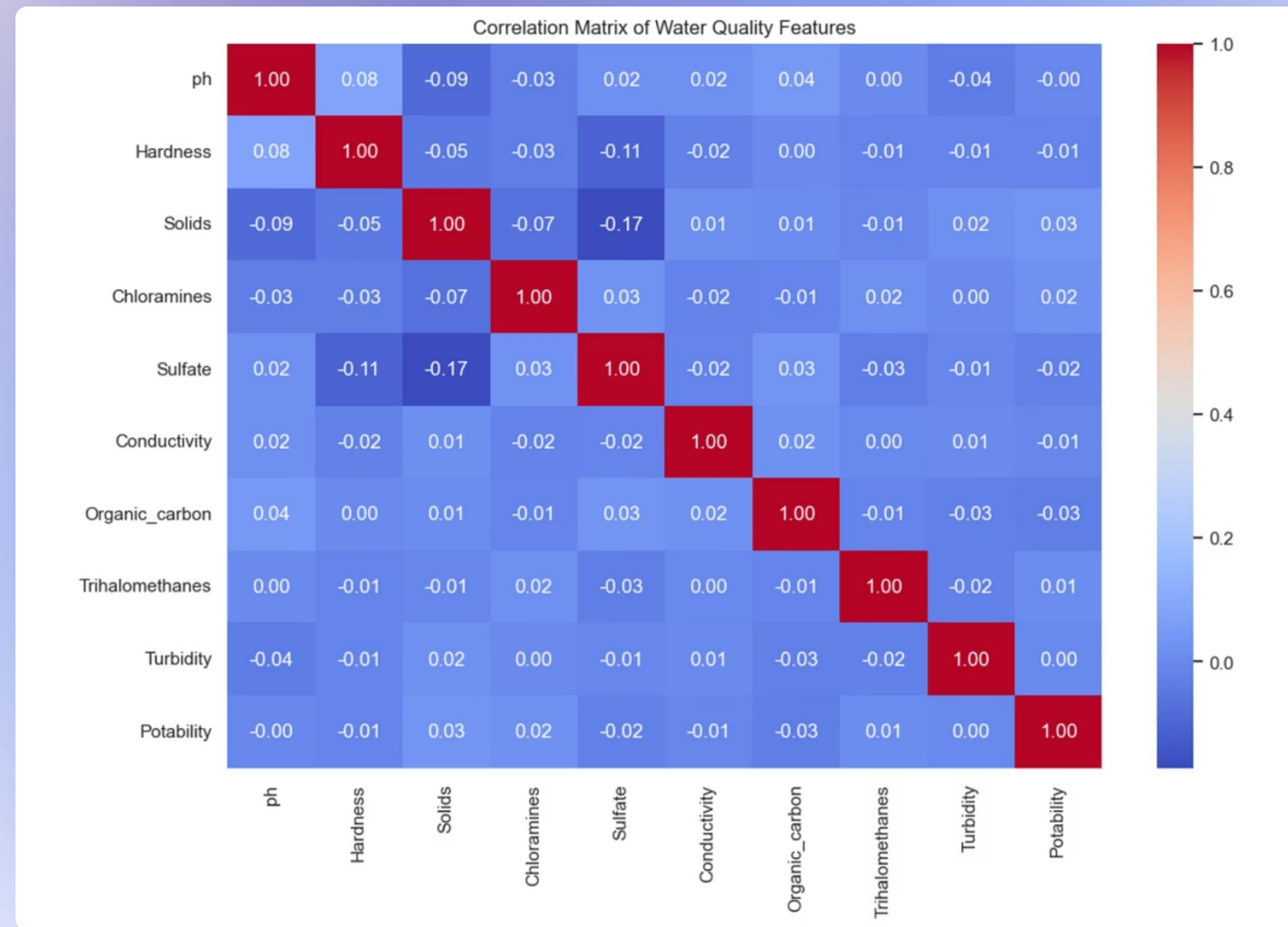
Exploratory Data Analysis (EDA)

Correlation Analysis

Heatmap visualization reveals low linear correlation between individual features.

Insight

No single chemical parameter guarantees safety, necessitating a non-linear Machine Learning approach (Random Forest).



Outlier Detection

Method

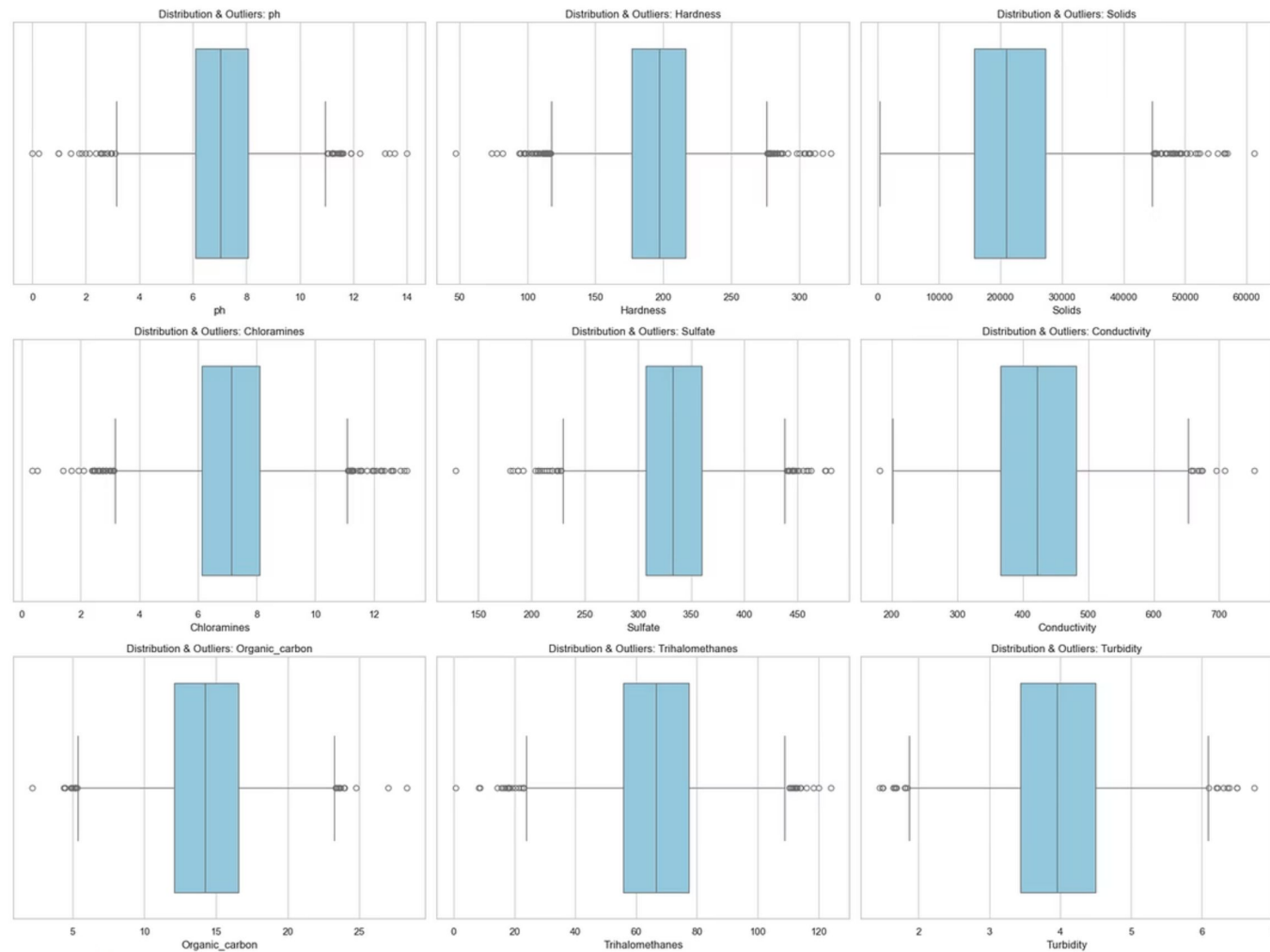
Univariate analysis using Boxplots.

Observation

Significant outliers detected in Solids and Conductivity.

Decision

Outliers were retained as they represent real-world chemical variations rather than sensor errors.



```
Potability
0    0.60989
1    0.39011
Name: proportion, dtype: float64
```

Model Architecture: Random Forest Classifier

Algorithm

Random Forest (Ensemble Learning)

Hyperparameters

`n_estimators=200, Random State=42.`

Why this model?

- Handles non-linear relationships between chemical parameters effectively.
- Robust against overfitting compared to a single Decision Tree.
- Provides feature importance rankings.

Model Evaluation (Metrics)

Classification Problem: Predicting Discrete Classes (Safe vs. Unsafe)



Accuracy

Overall correctness of the model.



Precision & Recall

Minimizing False Positives
(classifying unsafe water as
safe).

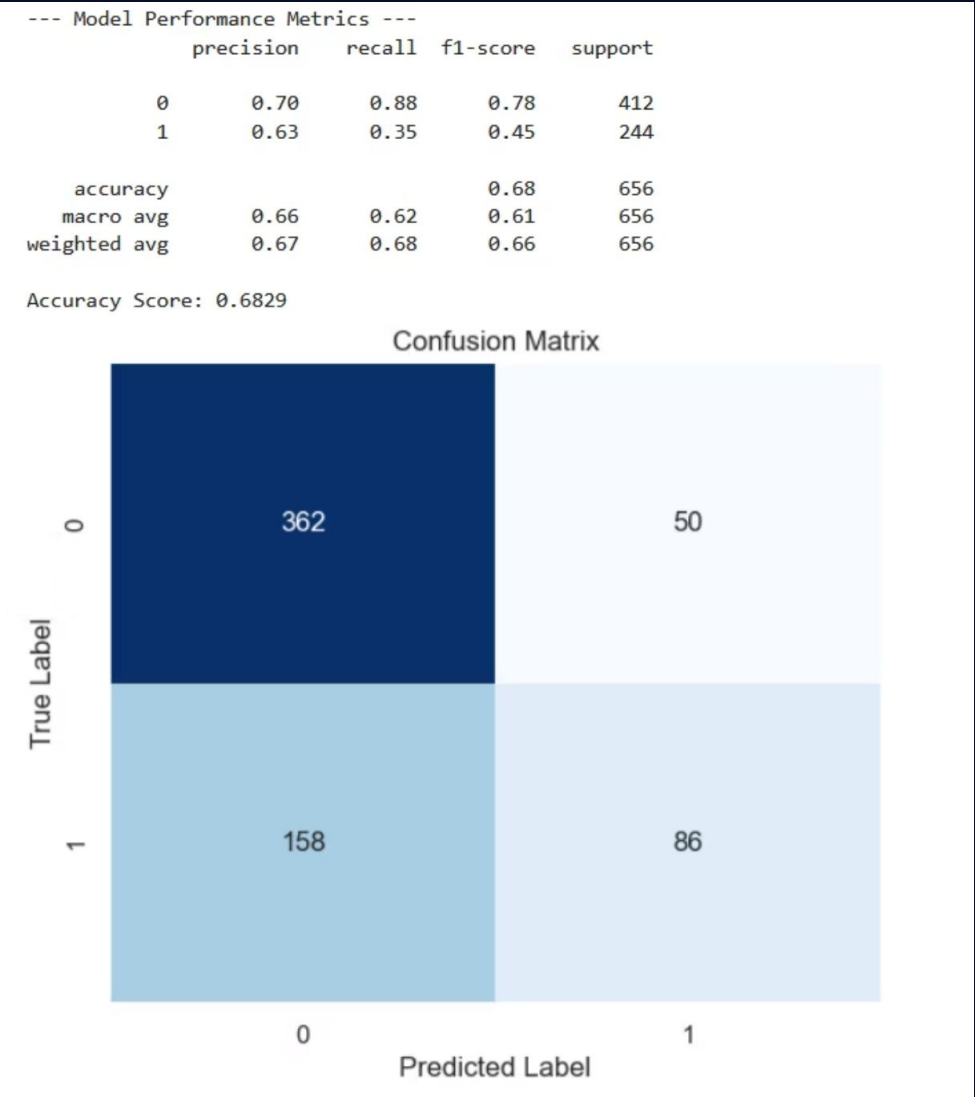


F1-Score

Harmonic mean of Precision and
Recall.

Key Metrics used instead of Regression metrics.

Results: Confusion Matrix & ROC Curve



Analysis

The Confusion Matrix highlights the False Negative vs. False Positive rate.



ROC Curve

Area Under Curve (AUC) demonstrates the model's ability to distinguish between potable and non-potable samples better than random guessing.

Cross-Validation Strategy



Methodology

Stratified K-Fold Cross-Validation
(k=5).



Purpose

To ensure the model performance
is consistent across different
subsets of the data and not just
lucky on one test split.



Result

Consistent accuracy scores
across all 5 folds confirm model
stability.

```
Cross-Validation Scores (5 Folds): [0.66920732 0.69312977 0.67022901 0.65954198 0.66412214]  
Mean CV Accuracy: 0.6712
```